ISYE 6414 - Life Expectancy Regression Analysis

Introduction

The objective of this project is to identify which factors have the largest marginal association with life expectancy across countries. The analysis uses a country-year panel dataset constructed from three sources: the Human Development Index dataset, World Development Indicators, and World Health Organization data. The analysis focuses on regression modeling, exploratory data analysis, variable selection, diagnostics, validation, and interpretation of final model results.

Data sources:

World Health Organization (WHO): 24,000+ observations from 2000-2021

World Development Indicators (WDI): 10,000+ observations from 2000-2025

UNDP (HDI): 1000+ observations, 206 countries from 1990-2023

Loading Libraries

In [464…

```
library(data.table)
library(dplyr)
library(tidyr)
library(MASS)
library(car)
library(lmtest)
library(sandwich)
library(ggplot2)
library(caret)
set.seed(100)
```

Importing the Datasets and Basic Data Exploration

In [465…

```
hdi_base <- read.csv("HDI Dataset.csv", header = TRUE)
wdi_base <- read.csv("WDI Dataset.csv", header = TRUE)
who_base <- read.csv("WHO Dataset.csv", header = TRUE)

names(hdi_base) <- tolower(names(hdi_base))
```

```
names(wdi_base) <- tolower(names(wdi_base))
names(who_base) <- tolower(names(who_base))
```

In [466…

```
head(hdi_base)
```

	iso3	country	hdicode	region	hdi_rank_2023	hdi_1990	hdi_1991	hdi_1992	hdi_1
	<chr>	<chr>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>	<d
1	AFG	Afghanistan	Low	SA	181	0.285	0.291	0.301	0.
2	ALB	Albania	Very High	ECA	71	0.654	0.638	0.622	0.
3	DZA	Algeria	High	AS	96	0.595	0.596	0.601	0.
4	AND	Andorra	Very High		32	NA	NA	NA	
5	AGO	Angola	Medium	SSA	148	NA	NA	NA	
6	ATG	Antigua and Barbuda	Very High	LAC	53	NA	NA	NA	

In [467…

```
head(wdi_base)
```

	country.name	country.code	series.name	series.code	x2000..yr2000.	x200
	<chr>	<chr>	<chr>	<chr>	<chr>	
1	Afghanistan	AFG	GDP per capita (current US$)	NY.GDP.PCAP.CD	174.930991430166	138.7068
2	Afghanistan	AFG	GDP per capita growth (annual %)	NY.GDP.PCAP.KD.ZG	..	-10.11948
3	Afghanistan	AFG	Inflation, GDP deflator (annual %)	NY.GDP.DEFL.KD.ZG	..	-11.77453
4	Afghanistan	AFG	Life expectancy at birth, total (years)	SP.DYN.LE00.IN	55.005	
5	Afghanistan	AFG	Population growth (annual %)	SP.POP.GROW	1.21217597601851	0.7620048
6	Afghanistan	AFG	Population, male	SP.POP.TOTL.MA.IN	10094645	

In [468…

```
head(who_base)
```

	indicatorcode	indicator	valuetype	parentlocationcode	parentlocation	location.type
	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
1	WHOSIS_000001	Life expectancy at birth (years)	text	AFR	Africa	Country
2	WHOSIS_000001	Life expectancy at birth (years)	text	AFR	Africa	Country
3	WHOSIS_000001	Life expectancy at birth (years)	text	AFR	Africa	Country
4	WHOSIS_000001	Life expectancy at birth (years)	text	AFR	Africa	Country
5	WHOSIS_000001	Life expectancy at birth (years)	text	EMR	Eastern Mediterranean	Country
6	WHOSIS_000001	Life expectancy at birth (years)	text	AFR	Africa	Country

---

Data Cleaning and Transformation

The HDI data is reshaped from wide format to long format and then widened again into one row per country-year. This keeps the same preprocessing structure as the initial analysis while making the dataset usable for regression modeling.

In [469…

```
hdi_subset <- hdi_base %>% dplyr::select(iso3, country, region,
                                         starts_with("hdi_"), starts_with("le_"),
                                         starts_with("eys_"), starts_with("mys_"),
                                         starts_with("gnipc_"))
hdi_long <- hdi_subset %>%
pivot_longer(cols = -c(iso3, country, region), names_to="variable_year", values_to=
hdi_long <- hdi_long %>%
separate(variable_year, into = c("variable", "year"), sep= "_(?=[^_]+$)") %>%
mutate(year=as.numeric(year))
hdi_clean <- hdi_long %>%
```

```
pivot_wider(names_from= variable, values_from=value)
head(hdi_clean)
dim(hdi_clean)
names(hdi_clean)
```

A tibble: 6 × 18

iso3	country	region	year	hdi_rank	hdi	hdi_f	hdi_m	le	le_f
<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
AFG	Afghanistan	SA	2023	181	0.496	0.3792537	0.5746028	66.035	67.536
AFG	Afghanistan	SA	1990	NA	0.285	NA	NA	45.118	47.703
AFG	Afghanistan	SA	1991	NA	0.291	NA	NA	45.521	48.282
AFG	Afghanistan	SA	1992	NA	0.301	NA	NA	46.569	49.607
AFG	Afghanistan	SA	1993	NA	0.311	NA	NA	51.021	52.296
AFG	Afghanistan	SA	1994	NA	0.305	NA	NA	50.969	53.135

7004 · 18

'iso3' · 'country' · 'region' · 'year' · 'hdi_rank' · 'hdi' · 'hdi_f' · 'hdi_m' · 'le' · 'le_f' · 'le_m' · 'eys' · 'eys_f' · 'eys_m' · 'mys' · 'mys_f' · 'mys_m' · 'gnipc'

The WHO data is already close to long format, so it is summarized by country, year, and indicator, then converted to wide format.

In [470…

```
who_subset <- who_base %>% dplyr::select(iso3=spatialdimvaluecode,
                                         country=location, year = period,
                                         indicator=indicator, value=factvaluenumeri
who_subset$year <- as.numeric(who_subset$year)
who_clean <- who_subset %>%
group_by(iso3, country, year, indicator) %>%
summarise(value=mean(value, na.rm = TRUE), .groups="drop")
who_clean <- who_clean %>%
pivot_wider(names_from=indicator, values_from=value)
names(who_clean) <- make.names(names(who_clean))
head(who_clean)
dim(who_clean)
names(who_clean)
```

A tibble: 6 × 5

iso3	country	year	Life.expectancy.at.age.60..years.	Life.expectancy.at.birth..years.
<chr>	<chr>	<dbl>	<dbl>	<dbl>
AFG	Afghanistan	2000	13.14000	53.83333
AFG	Afghanistan	2001	13.13333	53.92000
AFG	Afghanistan	2002	13.22000	55.15667
AFG	Afghanistan	2003	13.40667	56.09333
AFG	Afghanistan	2004	13.53667	56.48667
AFG	Afghanistan	2005	13.65333	56.80000

4070 · 5

'iso3' · 'country' · 'year' · 'Life.expectancy.at.age.60..years.' · 'Life.expectancy.at.birth..years.'

The WDI data is filtered to variables relevant to economic conditions, population, education, health spending, survival, and mortality. It is then converted from wide year columns to long format and widened into one country-year row.

In [471…

```
wdi_dt <- as.data.table(wdi_base)
keep_vars <- c("GDP per capita (current US$)",
               "GDP per capita growth (annual %)",
               "Inflation, GDP deflator (annual %)",
               "Population growth (annual %)",
               "School enrollment, secondary (% gross)",
               "Out-of-pocket expenditure per capita, PPP (current international $)
               "Survival to age 65, male (% of cohort)",
               "Survival to age 65, female (% of cohort)",
               "Mortality rate, infant, male (per 1,000 live births)",
               "Mortality rate, infant, female (per 1,000 live births)",
               "Mortality rate, adult, male (per 1,000 male adults)",
               "Mortality rate, adult, female (per 1,000 female adults)")
wdi_dt <- wdi_dt[series.name %in% keep_vars]

wdi_long <- melt(wdi_dt, id.vars=c("country.code", "country.name", "series.name"),
                 measure.vars = patterns("^x"), variable.name = "year", value.name
wdi_long[, year := as.numeric(gsub("x|\\.\\..*", "", year))]

setnames(wdi_long,
         c("country.code", "country.name", "series.name"),
         c("iso3", "country", "indicator"))
wdi_long[, value := suppressWarnings(as.numeric(value))]
wdi_long <- wdi_long[, .(value=base::mean(value, na.rm = TRUE)), by= .(iso3, countr

wdi_clean <- dcast(wdi_long, iso3 + country + year ~ indicator, value.var="value")
names(wdi_clean) <- make.names(names(wdi_clean))
setorder(wdi_clean, iso3, year)
wdi_clean[] <- lapply(wdi_clean, function(x) {if (is.numeric(x)) {x[is.nan(x)] <- N
dim(wdi_clean)
```

```
head(wdi_clean)
names(wdi_clean)
```

6916 · 15

iso3	country	year	GDP.per.capita..current.US..	GDP.per.capita.growth..annual...	Inflation
<chr>	<chr>	<dbl>	<dbl>	<dbl>	
ABW	Aruba	2000	20681.02	6.51922377	
ABW	Aruba	2001	20740.13	3.21240569	
ABW	Aruba	2002	21307.25	-1.62809915	
ABW	Aruba	2003	21949.49	-0.03383944	
ABW	Aruba	2004	23700.63	5.02691152	
ABW	Aruba	2005	24171.84	-2.93082349	

'iso3' · 'country' · 'year' · 'GDP.per.capita..current.US..' · 'GDP.per.capita.growth..annual...' · 'Inflation..GDP.deflator..annual...' · 'Mortality.rate..adult..female..per.1.000.female.adults.' · 'Mortality.rate..adult..male..per.1.000.male.adults.' · 'Mortality.rate..infant..female..per.1.000.live.births.' · 'Mortality.rate..infant..male..per.1.000.live.births.' · 'Out.of.pocket.expenditure.per.capita..PPP..current.international...' · 'Population.growth..annual...' · 'School.enrollment..secondary....gross.' · 'Survival.to.age.65..female....of.cohort.' · 'Survival.to.age.65..male....of.cohort.'

Data Integration and Panel Construction

I merged the three datasets by ISO3 country code and year. The response variable is set to life expectancy from the HDI data.

In [472...

```
merged_data <- hdi_clean %>%
dplyr::full_join(who_clean, by=c("iso3", "year")) %>% # adding WHO
dplyr::full_join(wdi_clean, by=c("iso3", "year")) # adding WDI
merged_data <- merged_data %>%
dplyr::select(-country.y, -country) %>% # removing duplicate country columns
dplyr::rename(country = country.x) %>% # keep it to 1

dplyr::select(-dplyr::contains("Life.expectancy")) %>%
dplyr::select(-hdi_f, -hdi_m, -le_f, -le_m, -eys_f, -eys_m, -mys_f, -mys_m) %>%

# Creating variables
dplyr::mutate(life_exp=le,
              log_gdp=log(`GDP.per.capita..current.US..` + 1),
              health_spending=`Out.of.pocket.expenditure.per.capita..PPP..current.i
              population_growth=`Population.growth..annual...`,
              school_enrollment=`School.enrollment..secondary....gross.`,
```

```
                # Mortality variables
                mort_inf_f=`Mortality.rate..infant..female..per.1.000.live.births.`,
                mort_inf_m=`Mortality.rate..infant..male..per.1.000.live.births.`,
                mort_adult_f=`Mortality.rate..adult..female..per.1.000.female.adults.
                mort_adult_m=`Mortality.rate..adult..male..per.1.000.male.adults.`,
                # Survival rates to age 65
                survival65_f=`Survival.to.age.65..female....of.cohort.`,
                survival65_m=`Survival.to.age.65..male....of.cohort.`)
```

Final Cleaning and Modeling Dataset

The merged dataset is filtered to the 2000 - 2023 period used in this analysis. The goal here is to keep valid country-year observations and prepare a complete dataset for regression modeling.

In [473…

```
panel_clean <- merged_data %>%
dplyr::filter(year >= 2000, year <= 2023, !is.na(life_exp), !is.na(country), !is.na

# Fills missing numeric values using the country median or overall median if needed
num_cols <- names(panel_clean)[sapply(panel_clean, is.numeric)]
for (j in num_cols) {panel_clean <- panel_clean %>%
    group_by(iso3) %>%
    mutate("{j}":=ifelse(is.na(.data[[j]]), median(.data[[j]], na.rm = TRUE), .data
    ungroup()
    panel_clean[[j]][is.nan(panel_clean[[j]])] <- NA
    panel_clean[[j]][is.na(panel_clean[[j]])] <- median(panel_clean[[j]],na.rm = TR

# Removes infinite values from log transformations or conversions
panel_clean <- panel_clean %>%
mutate(across(where(is.numeric),~ifelse(is.infinite(.), NA, .))) %>%
na.omit()

cat("Dimensions:\n")
print(dim(panel_clean))

cat("\nVariable names:\n")
print(names(panel_clean))

cat("\nTotal Missing:\n")
print(sum(colMeans(is.na(panel_clean))))
```

```
Dimensions:
[1] 4944   33

Variable names:
 [1] "iso3"
 [2] "country"
 [3] "region"
 [4] "year"
 [5] "hdi_rank"
 [6] "hdi"
 [7] "le"
 [8] "eys"
 [9] "mys"
[10] "gnipc"
[11] "GDP.per.capita..current.US.."
[12] "GDP.per.capita.growth..annual..."
[13] "Inflation..GDP.deflator..annual..."
[14] "Mortality.rate..adult..female..per.1.000.female.adults."
[15] "Mortality.rate..adult..male..per.1.000.male.adults."
[16] "Mortality.rate..infant..female..per.1.000.live.births."
[17] "Mortality.rate..infant..male..per.1.000.live.births."
[18] "Out.of.pocket.expenditure.per.capita..PPP..current.international..."
[19] "Population.growth..annual..."
[20] "School.enrollment..secondary....gross."
[21] "Survival.to.age.65..female....of.cohort."
[22] "Survival.to.age.65..male....of.cohort."
[23] "life_exp"
[24] "log_gdp"
[25] "health_spending"
[26] "population_growth"
[27] "school_enrollment"
[28] "mort_inf_f"
[29] "mort_inf_m"
[30] "mort_adult_f"
[31] "mort_adult_m"
[32] "survival65_f"
[33] "survival65_m"

Total Missing:
[1] 0
```

After filtering and imputation, the final modeling dataset contains 4,944 country-year observations and 33 variables. The missing value check shows that all variables have 0% missing, meaning the dataset is ready for regression modeling.

In [474…

```
head(panel_clean)
dim(panel_clean)
```

iso3	country	region	year	hdi_rank	hdi	le	eys	mys	gnipc
<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
AFG	Afghanistan	SA	2023	181	0.496	66.035	10.790143	2.514790	1971.549
AFG	Afghanistan	SA	2000	181	0.351	55.005	5.856422	1.264052	1331.140
AFG	Afghanistan	SA	2001	181	0.355	55.511	6.148418	1.315551	1246.158
AFG	Afghanistan	SA	2002	181	0.383	56.225	6.440414	1.367049	1946.696
AFG	Afghanistan	SA	2003	181	0.392	57.171	6.732410	1.418548	1982.540
AFG	Afghanistan	SA	2004	181	0.408	57.810	7.600430	1.470046	1963.735

4944 · 33

Core identifiers

- region: geographic grouping
- year: observation year

Development / Human capital

- hdi: composite index (health + education + income)
- hdi_rank: rank based on HDI
- eys: expected years of schooling
- mys: mean years of schooling
- school_enrollment: % enrolled in school

Economic variables

- gnipc: gross national income per capita
- GDP.per.capita..current.US..: GDP per person
- GDP.per.capita.growth..annual...: GDP growth rate
- log_gdp: log-transformed GDP (used for modeling)

Health variables

- life_exp / le: life expectancy (your response variable)
- Mortality.rate..adult..female...
- Mortality.rate..adult..male...
- Mortality.rate..infant..female...
- Mortality.rate..infant..male...
- Survival.to.age.65..female...
- Survival.to.age.65..male...: % surviving to 65
- Out.of.pocket.expenditure...: healthcare spending by individuals

- health_spending

Demographics

- Population.growth..annual...
- population_growth

Additional Macro conditions

- Inflation..GDP.deflator..annual...

Variable Selections and Analysis Dataset

This merged dataset keeps variables related to income, education, health spending, mortality, survival, and region. HDI is kept for the separate human capital model, but the final variable selection model avoids using HDI because it already includes life expectancy as part of its index.

In [475...

```
model_data <- panel_clean %>%
dplyr::select(iso3, country, region, year,
              life_exp, hdi, eys, mys, gnipc,
              log_gdp, GDP.per.capita.growth..annual..., Inflation..GDP.deflator..a
              health_spending, population_growth, school_enrollment,
              mort_inf_f, mort_inf_m, mort_adult_f, mort_adult_m,
              survival65_f, survival65_m)

# reclassified countries into correct regions
eca_countries <- c("Andorra","Austria","Belgium","Bulgaria","Croatia","Cyprus","Cze
"Germany","Greece","Hungary","Iceland","Ireland", "Israel", "Italy","Latvia","Liech
"Malta","Monaco","Netherlands","Norway","Poland","Portugal","Romania","Russian Fede
"Slovakia","Slovenia","Spain","Sweden","Switzerland","United Kingdom")

eap_countries <- c("Australia","Japan","Korea (Republic of)","New Zealand","Hong Ko

lac_countries <- c("Canada","United States")

non_countries <- c("East Asia and the Pacific", "Europe and Central Asia",
                   "Latin America and the Caribbean", "South Asia",
                   "Sub-Saharan Africa", "World",
                   "Very high human development", "High human development",
                   "Medium human development", "Low human development", "Arab State

model_data <- model_data %>%
    dplyr::filter(!country %in% non_countries) %>%
    dplyr::mutate(region=
        ifelse(country %in% eca_countries, "ECA",
        ifelse(country %in% eap_countries, "EAP",
        ifelse(country %in% lac_countries, "LAC", region))))

# region as factor
model_data$region <- as.factor(model_data$region)
```

```
dim(model_data)
summary(model_data$life_exp)
```

4680 · 21

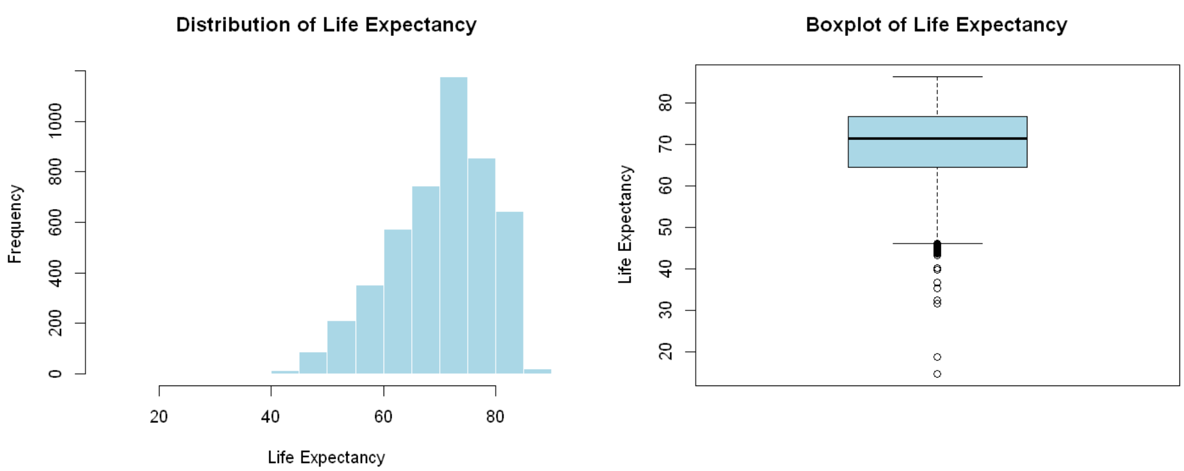
```
 Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
14.66   64.49   71.55   70.14   76.77   86.37 
```

Exploratory Data Analysis

In [476…

```
options(repr.plot.width = 12, repr.plot.height = 5)
par(mfrow = c(1,2))

# Distribution of life expectancy (response)
hist(model_data$life_exp, main="Distribution of Life Expectancy",
     xlab="Life Expectancy", col="Lightblue", border="white")

# Boxplot helps show outliers
boxplot(model_data$life_exp, main="Boxplot of Life Expectancy",
        ylab="Life Expectancy", col="Lightblue")

# Summary statistics for the response
summary(model_data$life_exp)
```

```
 Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
14.66   64.49   71.55   70.14   76.77   86.37 
```

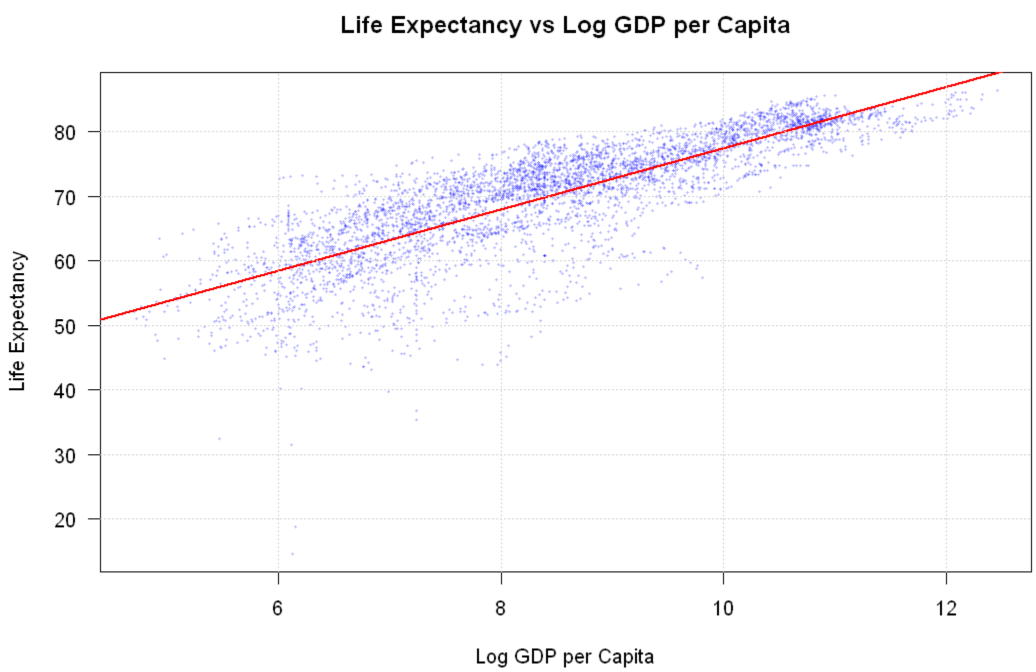


The distribution of life expectancy is slightly left skewed, with most countries clustered between 65 and 80 years. The boxplot highlights several lower expectancy outliers. Summary statistics show a mean of approximately 70 years and a median of 71.5 years, with a wide overall range. These results suggest that while most countries have relatively high life expectancy, a few of the countries experience lower outcomes.

In [477…

```
options(repr.plot.width = 9, repr.plot.height = 6)

# Life expectancy compared to income
plot(model_data$log_gdp, model_data$life_exp,
```

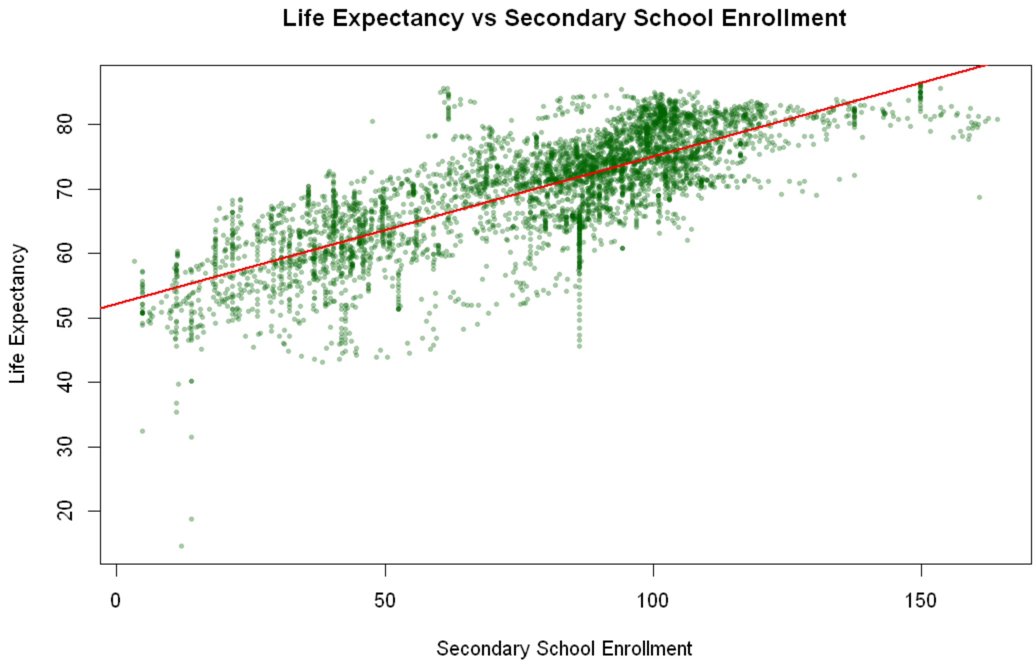
```
     main="Life Expectancy vs Log GDP per Capita",
     xlab="Log GDP per Capita",
     ylab="Life Expectancy",

     pch=16, cex=0.35, col=rgb(0, 0, 1, 0.25), las = 1)

abline(lm(life_exp ~ log_gdp, data=model_data), col="red", lwd = 2)

grid()
```



There is a clear positive relationship between log GDP per capita and life expectancy. Countries with higher income levels tend to have higher life expectancy. The relationship appears approximately linear after the log transformation, and greater variability is observed among lower income countries, while higher income countries are more clustered.

In [478…

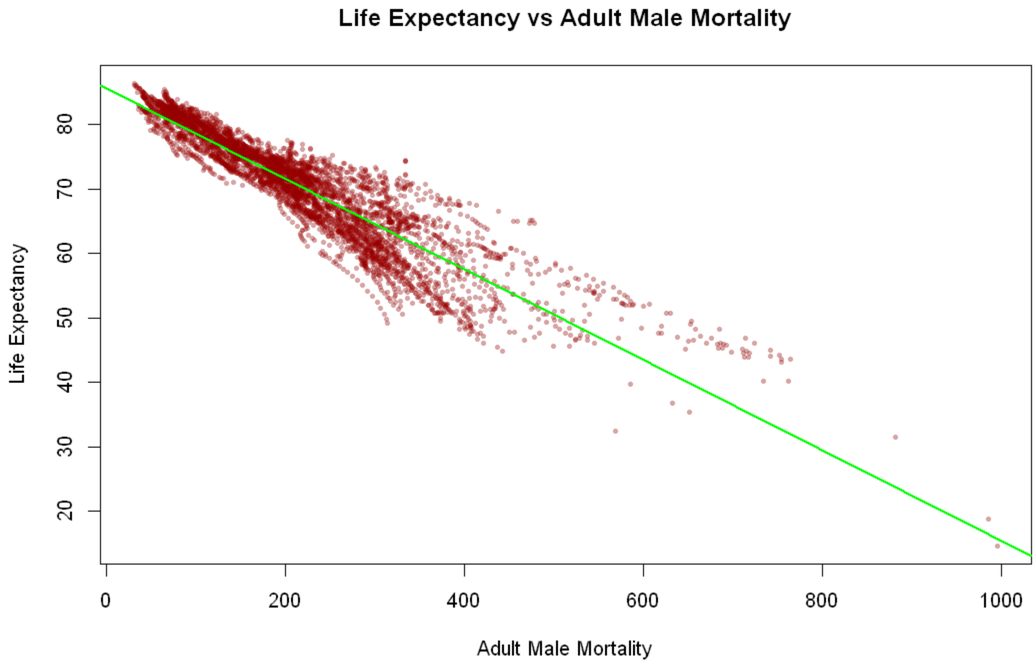
```
options(repr.plot.width = 9, repr.plot.height = 6)

# Life expectancy compared to education
plot(model_data$school_enrollment, model_data$life_exp,

     main="Life Expectancy vs Secondary School Enrollment",
     xlab="Secondary School Enrollment", ylab="Life Expectancy",

     pch=16, cex=0.5, col=rgb(0,0.4,0,0.35))

abline(lm(life_exp ~ school_enrollment, data=model_data), col="red", lwd = 2)
```



There is a clear positive relationship between secondary school enrollment and life expectancy. Countries with higher enrollment rates tend to have higher life expectancy. The relationship appears approximately linear, and greater variability is observed at lower enrollment levels, while higher enrollment levels are more clustered.

In [479…

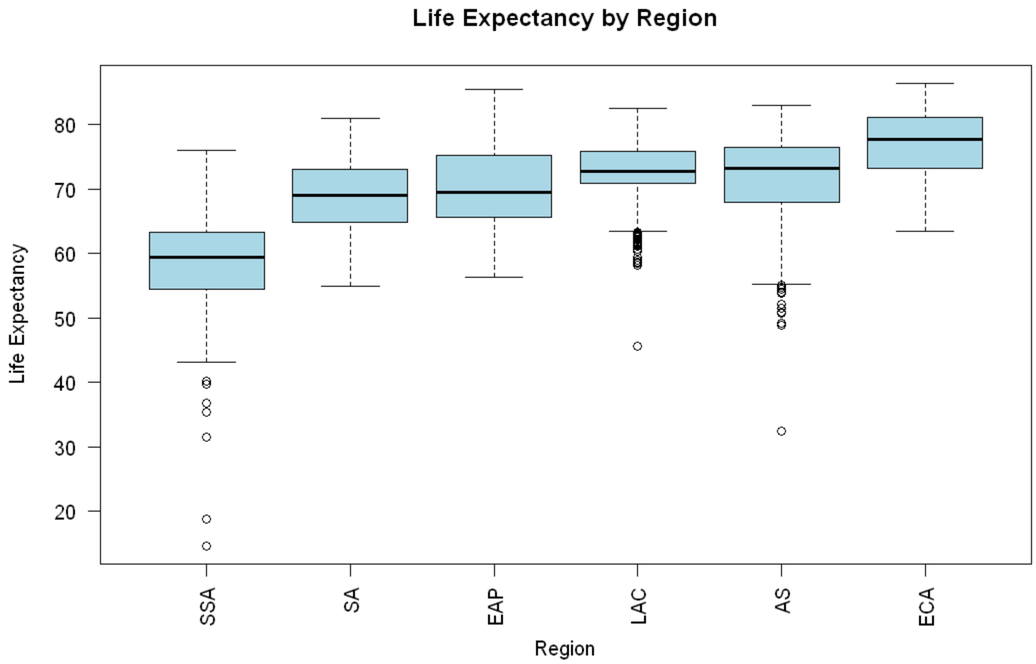
```
options(repr.plot.width = 9, repr.plot.height = 6)

# Life expectancy compared to adult mortality
plot(model_data$mort_adult_m, model_data$life_exp,

     main="Life Expectancy vs Adult Male Mortality",
     xlab="Adult Male Mortality", ylab="Life Expectancy",

     pch=16, cex=0.5, col=rgb(0.6,0,0,0.35))

abline(lm(life_exp ~ mort_adult_m, data=model_data), col="green", lwd = 2)
```



There is a strong negative relationship between adult male mortality and life expectancy. Countries with higher adult mortality rates tend to have lower life expectancy. The relationship appears approximately linear, with higher mortality levels show greater variability and lower mortality levels are more clustered.

In [480…

```
# Regional differences in life expectancy
model_data_boxplot <- model_data
model_data_boxplot$region <- reorder(model_data_boxplot$region, model_data_boxplot$
boxplot(life_exp ~ region, data=model_data_boxplot,
        main="Life Expectancy by Region",
        xlab="Region", ylab="Life Expectancy", col="light blue",
        las = 2)
```

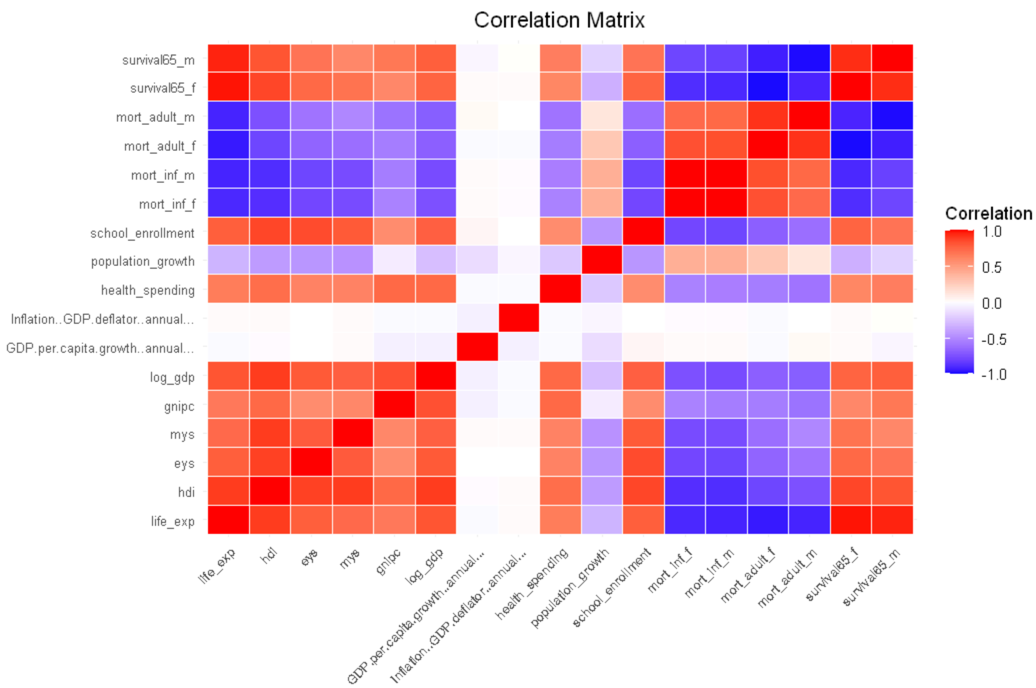


The boxplot shows variation in life expectancy across regions. Europe and Central Asia (ECA) has the highest median life expectancy, followed by Latin America (LAC) and East Asia and Pacific (EAP). South Asia (SA) and the Middle East (SAS) fall in the middle range, while Sub-Saharan Africa (SSA) has the lowest median and the widest spread.

SSA also exhibits the greatest variability and several low outliers, indicating significant disparities within the region. Higher performing regions such as ECA show more clustered distributions, suggesting more consistent life expectancy outcomes.

In [481…

```
# Correlation matrix using ggplot
# Select numeric variables
correlation1 <- model_data %>%
dplyr::select(where(is.numeric)) %>%
dplyr::select(-year)
correlation1 <- correlation1[, apply(correlation1, 2, var, na.rm=TRUE) !=0] # Remov

corr_mat <- cor(correlation1, use="complete.obs")
corr_mat[is.na(corr_mat)] <- 0

corr_df <- as.data.frame(as.table(corr_mat))
names(corr_df) <- c("Variable1", "Variable2", "Correlation")

ggplot(corr_df, aes(x=Variable1, y=Variable2, fill=Correlation)) +
    geom_tile(color="white") +
    scale_fill_gradient2(low="blue", high="red", mid="white", midpoint=0, limit= c(
    labs(title="Correlation Matrix", x="", y="", fill="Correlation") +
    theme_minimal() +
    theme(axis.text.x=element_text(angle=45, hjust=1, size=8),
          axis.text.y=element_text(size=8),
          plot.title=element_text(hjust=0.5))
```



The correlation heatmap shows strong positive relationships between life expectancy and development indicators such as HDI, education, and income. Mortality variables are strongly negatively correlated with life expectancy and positively correlated with each other. These patterns suggest potential multicollinearity

---

Base Model & Diagnostics

In [482...

```
base_model <- lm(life_exp ~ hdi + eys + mys + log_gdp +
                 GDP.per.capita.growth..annual... + Inflation..GDP.deflator..annual
                 health_spending + population_growth + school_enrollment +
                 mort_inf_f + mort_inf_m + mort_adult_f + mort_adult_m +
                 survival65_f + survival65_m, data=model_data)


vif_df <- data.frame(Variable=names(vif(base_model)), VIF=vif(base_model)) %>%
arrange(desc(VIF))

vif_df
summary(base_model)
```

A data.frame: 15 × 2

	Variable	VIF
	<chr>	<dbl>
survival65_f	survival65_f	451.831281
mort_adult_f	mort_adult_f	288.943534
survival65_m	survival65_m	260.765699
mort_inf_m	mort_inf_m	255.995839
mort_inf_f	mort_inf_f	224.415797
mort_adult_m	mort_adult_m	191.674910
hdi	hdi	37.954525
mys	mys	8.508357
log_gdp	log_gdp	8.145774
eys	eys	6.518610
school_enrollment	school_enrollment	5.032523
health_spending	health_spending	2.492542
population_growth	population_growth	1.663901
GDP.per.capita.growth..annual...	GDP.per.capita.growth..annual...	1.047867
Inflation..GDP.deflator..annual...	Inflation..GDP.deflator..annual...	1.007983

```
Call:
lm(formula = life_exp ~ hdi + eys + mys + log_gdp + GDP.per.capita.growth..annual...
+
    Inflation..GDP.deflator..annual... + health_spending + population_growth +
    school_enrollment + mort_inf_f + mort_inf_m + mort_adult_f +
    mort_adult_m + survival65_f + survival65_m, data = model_data)

Residuals:
     Min       1Q   Median       3Q      Max
-14.5990  -0.5050  -0.0606   0.4025   3.4843

Coefficients:
                                     Estimate Std. Error t value Pr(>|t|)
(Intercept)                         9.113e+00  8.660e-01  10.524  < 2e-16 ***
hdi                                 2.577e-01  4.486e-01   0.575  0.56565
eys                                 7.226e-02  9.097e-03   7.944 2.44e-15 ***
mys                                 5.392e-02  1.039e-02   5.191 2.19e-07 ***
log_gdp                             1.531e-01  2.150e-02   7.124 1.21e-12 ***
GDP.per.capita.growth..annual...    3.258e-03  2.142e-03   1.521  0.12845
Inflation..GDP.deflator..annual... -1.845e-06  3.525e-06  -0.523  0.60068
health_spending                     4.418e-04  5.947e-05   7.429 1.29e-13 ***
population_growth                  -2.567e-02  9.445e-03  -2.717  0.00661 **
school_enrollment                  -3.576e-03  8.590e-04  -4.164 3.19e-05 ***
mort_inf_f                         -9.817e-02  7.549e-03 -13.005  < 2e-16 ***
mort_inf_m                          5.143e-02  6.827e-03   7.534 5.87e-14 ***
mort_adult_f                        2.961e-02  1.865e-03  15.872  < 2e-16 ***
mort_adult_m                        1.685e-03  1.385e-03   1.216  0.22404
survival65_f                        4.991e-01  1.687e-02  29.581  < 2e-16 ***
survival65_m                        2.397e-01  1.247e-02  19.231  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7961 on 4664 degrees of freedom
Multiple R-squared:  0.992,	Adjusted R-squared:  0.992
F-statistic: 3.876e+04 on 15 and 4664 DF,  p-value: < 2.2e-16
```

Initial Summary and VIF

I first ran a base model including all selected variables to see overall relationships with life expectancy.

The model has a very high R-squared (~0.992), which means it explains most of the variation in life expectancy. Most variables are statistically significant, so they do have some relationship with the response.

However, the VIF results show extremely high values for several variables, especially survival and mortality measures. This indicates strong multicollinearity, meaning many variables are overlapping and explaining the same thing.

I also included HDI along with education variables (eys, mys), which likely adds to the multicollinearity since HDI already captures similar information.

Overall, while the model fits well, the results are not reliable due to multicollinearity. We'll need to reduce variables further and improve model stability.

In [483...

```
cooks_d <- cooks.distance(base_model)
threshold <- 4 / nrow(model_data)
outliers <- which(cooks_d > threshold)
high_outliers <- which(cooks_d > 4*mean(cooks_d))
cat("Outliers:", length(outliers),"\n")
cat("High Outliers:", length(high_outliers),"\n")
```

```
Outliers: 250
High Outliers: 24
```

Outlier Screening with Cook's Distance

I used Cook's Distance to identify observations that may have a large influence on the regression model. Using the common threshold of 4 / n, 250 observations were flagged. Since removing all 250 would remove many valid country-year observations, I used a stricter rule and only removed observations with Cook's Distance greater than 4 * mean(cooks_d).

This resulted in 24 high influence observations being removed. This approach is more conservative because it removes only the most extreme influential points while keeping most of the country-level variation in the dataset.

In [484...

```
model_data_clean <- model_data[-high_outliers, ]

clean_model <- lm(life_exp ~ hdi + eys + mys +log_gdp +
                  GDP.per.capita.growth..annual... + Inflation..GDP.deflator..annua
                  health_spending + population_growth + school_enrollment +
                  mort_inf_f + mort_inf_m + mort_adult_f + mort_adult_m +
                  survival65_f + survival65_m,
                  data=model_data_clean)

summary(clean_model)
```

```
Call:
lm(formula = life_exp ~ hdi + eys + mys + log_gdp + GDP.per.capita.growth..annual...
+
    Inflation..GDP.deflator..annual... + health_spending + population_growth +
    school_enrollment + mort_inf_f + mort_inf_m + mort_adult_f +
    mort_adult_m + survival65_f + survival65_m, data = model_data_clean)

Residuals:
    Min      1Q  Median      3Q     Max
-3.2523 -0.4735 -0.0494  0.3910  2.9310

Coefficients:
                                     Estimate Std. Error t value Pr(>|t|)
(Intercept)                         2.371e+00  8.227e-01   2.882 0.003975 **
hdi                                 1.476e+00  4.277e-01   3.450 0.000566 ***
eys                                 5.763e-02  8.356e-03   6.897 6.03e-12 ***
mys                                 4.782e-02  9.414e-03   5.080 3.93e-07 ***
log_gdp                             8.362e-02  1.995e-02   4.192 2.82e-05 ***
GDP.per.capita.growth..annual...    3.627e-03  1.987e-03   1.825 0.067991 .
Inflation..GDP.deflator..annual... -2.947e-05  3.219e-05  -0.916 0.359975
health_spending                     3.943e-04  5.372e-05   7.341 2.50e-13 ***
population_growth                  -2.977e-02  8.477e-03  -3.512 0.000449 ***
school_enrollment                  -3.869e-03  7.698e-04  -5.027 5.18e-07 ***
mort_inf_f                         -6.139e-02  7.113e-03  -8.631  < 2e-16 ***
mort_inf_m                          3.075e-02  6.334e-03   4.854 1.25e-06 ***
mort_adult_f                        2.609e-02  1.734e-03  15.045  < 2e-16 ***
mort_adult_m                        1.282e-02  1.336e-03   9.590  < 2e-16 ***
survival65_f                        4.825e-01  1.561e-02  30.912  < 2e-16 ***
survival65_m                        3.264e-01  1.187e-02  27.483  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7113 on 4640 degrees of freedom
Multiple R-squared:  0.9935,    Adjusted R-squared:  0.9934
F-statistic: 4.7e+04 on 15 and 4640 DF,  p-value: < 2.2e-16
```

After removing the 24 high-influence observations, I refit the regression model. The cleaned model still has a very strong fit, with an adjusted R-squared of about 0.993, meaning the predictors explain most of the variation in life expectancy.

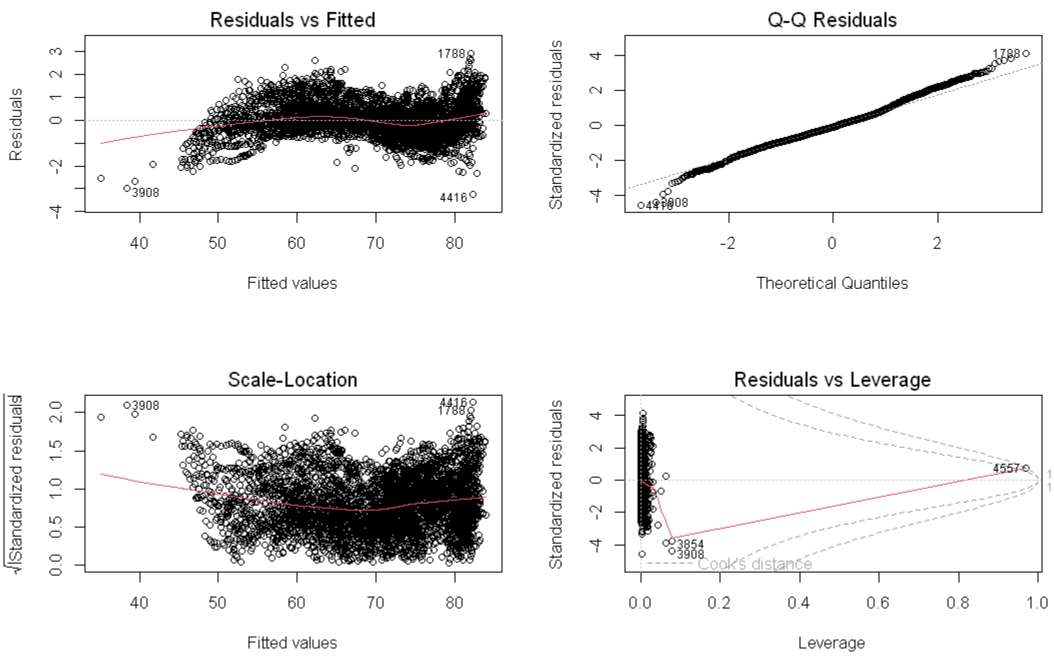
Most predictors remain statistically significant, including HDI, schooling variables, log GDP, health spending, mortality rates, and survival rates. This suggests that development, education, income, and health outcomes are all strongly related to life expectancy.

However, some coefficient signs are still not fully intuitive, especially because mortality and survival variables are highly related to each other. This suggests that multicollinearity is still affecting interpretation, so the next step should be variable selection as mentioned before.

Additional Diagnostics

In [485…

```
suppressWarnings({par(mfrow=c(2,2))
                  plot(clean_model)})
```



Residuals vs Fitted:

The residuals are mostly centered around zero, but there is some curvature in the pattern. This suggests that the relationship may not be perfectly linear and that some nonlinear effects could exist.

Q-Q Plot:

The points mostly follow the straight line, indicating that the residuals are approximately normally distributed. There are small deviations at the tails, but these are not severe and are common in real-world data.

Scale-Location Plot:

The spread of the residuals is not completely constant across fitted values. There is some increase in spread at higher fitted values, which suggests mild heteroskedasticity. However, the pattern is not extreme.

Residuals vs Leverage:

Most observations have low leverage, and only a few points stand out. The most extreme influential observations were already removed using Cook's Distance, so remaining points do not appear to overly distort the model.

In [486…

```
coeftest(clean_model, vcov=vcovHC(clean_model, type="HC1"))
```

```
t test of coefficients:

                                      Estimate  Std. Error t value  Pr(>|t|)
(Intercept)                         2.3706e+00  1.2308e+00  1.9261 0.0541509
hdi                                 1.4757e+00  6.1236e-01  2.4099 0.0159948
eys                                 5.7630e-02  1.0187e-02  5.6570 1.633e-08
mys                                 4.7820e-02  1.0436e-02  4.5824 4.718e-06
log_gdp                             8.3621e-02  2.5962e-02  3.2209 0.0012866
GDP.per.capita.growth..annual...    3.6273e-03  2.0078e-03  1.8066 0.0708898
Inflation..GDP.deflator..annual... -2.9473e-05  7.7433e-06 -3.8062 0.0001429
health_spending                     3.9434e-04  6.1552e-05  6.4066 1.635e-10
population_growth                  -2.9769e-02  1.0281e-02 -2.8956 0.0038023
school_enrollment                  -3.8694e-03  8.0974e-04 -4.7786 1.819e-06
mort_inf_f                         -6.1392e-02  1.0496e-02 -5.8492 5.279e-09
mort_inf_m                          3.0748e-02  8.6347e-03  3.5609 0.0003732
mort_adult_f                        2.6093e-02  2.3960e-03 10.8902 < 2.2e-16
mort_adult_m                        1.2815e-02  1.8658e-03  6.8684 7.350e-12
survival65_f                        4.8246e-01  2.0661e-02 23.3508 < 2.2e-16
survival65_m                        3.2635e-01  1.5869e-02 20.5656 < 2.2e-16

(Intercept)                        .
hdi                                *
eys                                ***
mys                                ***
log_gdp                            **
GDP.per.capita.growth..annual...   .
Inflation..GDP.deflator..annual... ***
health_spending                    ***
population_growth                  **
school_enrollment                  ***
mort_inf_f                         ***
mort_inf_m                         ***
mort_adult_f                       ***
mort_adult_m                       ***
survival65_f                       ***
survival65_m                       ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Robust Standard Errors

After applying robust standard errors, most variables remain statistically significant. This strengthens the analysis because it shows that the main relationships are still significant even after correcting for heteroskedasticity.

A few variables become weaker, such as GDP growth, but the major predictors remain important.

Removing variables with high multicollinearity

In [487…

```
reduced_model <- lm(life_exp ~ eys + mys + log_gdp + GDP.per.capita.growth..annual.
                    health_spending + population_growth + school_enrollment + mort_
```

```
                    mort_inf_f, data=model_data_clean)
summary(reduced_model)
```

```
Call:
lm(formula = life_exp ~ eys + mys + log_gdp + GDP.per.capita.growth..annual... +
    health_spending + population_growth + school_enrollment +
    mort_adult_f + mort_inf_f, data = model_data_clean)

Residuals:
    Min      1Q  Median      3Q     Max
-5.2077 -0.9607 -0.0246  0.9625  6.7438

Coefficients:
                                   Estimate Std. Error  t value Pr(>|t|)
(Intercept)                      69.3390016  0.2778482  249.557  < 2e-16 ***
eys                               0.1474971  0.0162670    9.067  < 2e-16 ***
mys                              -0.1029887  0.0142555   -7.224 5.85e-13 ***
log_gdp                           1.0018316  0.0329934   30.365  < 2e-16 ***
GDP.per.capita.growth..annual... -0.0102410  0.0043925   -2.331   0.0198 *
health_spending                   0.0019286  0.0001162   16.597  < 2e-16 ***
population_growth                 0.0741156  0.0176571    4.197 2.75e-05 ***
school_enrollment                 0.0003682  0.0016987    0.217   0.8284
mort_adult_f                     -0.0461459  0.0004172 -110.604  < 2e-16 ***
mort_inf_f                       -0.1062538  0.0024822  -42.806  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.586 on 4646 degrees of freedom
Multiple R-squared:  0.9674,    Adjusted R-squared:  0.9674
F-statistic: 1.533e+04 on 9 and 4646 DF,  p-value: < 2.2e-16
```

In [488…

```
vif_df_reduced <- data.frame(Variable=names(vif(reduced_model)), VIF=vif(reduced_mo
dplyr::arrange(desc(VIF))
vif_df_reduced
```

A data.frame: 9 × 2

	Variable	VIF
	<chr>	<dbl>
mort_inf_f	mort_inf_f	5.815635
eys	eys	5.161098
school_enrollment	school_enrollment	4.880552
log_gdp	log_gdp	4.752855
mys	mys	4.010968
mort_adult_f	mort_adult_f	3.530868
health_spending	health_spending	2.381854
population_growth	population_growth	1.455213
GDP.per.capita.growth..annual...	GDP.per.capita.growth..annual...	1.036203

The VIF values in the reduced model are significantly lower than before. All variables now have VIF values below approximately 6, which indicates that multicollinearity is no longer a major issue.

In [489…

```
# Variable selection using stepwise AIC
step_model <- stepAIC(reduced_model, direction= "both", trace=FALSE)
summary(step_model)
```

```
Call:
lm(formula = life_exp ~ eys + mys + log_gdp + GDP.per.capita.growth..annual... +
    health_spending + population_growth + mort_adult_f + mort_inf_f,
    data = model_data_clean)

Residuals:
    Min      1Q  Median      3Q     Max
-5.2081 -0.9599 -0.0259  0.9621  6.7442

Coefficients:
                                   Estimate Std. Error  t value Pr(>|t|)
(Intercept)                      69.3382934  0.2778005  249.597  < 2e-16 ***
eys                               0.1490162  0.0146788   10.152  < 2e-16 ***
mys                              -0.1022684  0.0138614   -7.378 1.89e-13 ***
log_gdp                           1.0028108  0.0326794   30.686  < 2e-16 ***
GDP.per.capita.growth..annual... -0.0101887  0.0043854   -2.323   0.0202 *
health_spending                   0.0019264  0.0001157   16.646  < 2e-16 ***
population_growth                 0.0736941  0.0175479    4.200 2.72e-05 ***
mort_adult_f                     -0.0461589  0.0004128 -111.810  < 2e-16 ***
mort_inf_f                       -0.1062953  0.0024745  -42.956  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.586 on 4647 degrees of freedom
Multiple R-squared:  0.9674,	Adjusted R-squared:  0.9674
F-statistic: 1.726e+04 on 8 and 4647 DF,  p-value: < 2.2e-16
```

In [490…

```
# Compare reduced model vs selected model
AIC(reduced_model, step_model)
BIC(reduced_model, step_model)
```

A data.frame: 2 × 2

	df	AIC
	<dbl>	<dbl>
reduced_model	11	17522.44
step_model	10	17520.49

A data.frame: 2 × 2

	df	BIC
	<dbl>	<dbl>
reduced_model	11	17593.35
step_model	10	17584.95

I applied stepwise regression using AIC to refine the reduced model. The procedure removed school_enrollment, indicating that it did not add meaningful explanatory power once other variables (like years of schooling and GDP) were already included.

The final model includes education (eys, mys), economic factors (log_gdp, GDP growth), health spending, population growth, and mortality variables. The model has a strong fit, with an adjusted R-squared of about 0.967, meaning it explains most of the variation in life expectancy even after simplifying.

Both AIC and BIC decreased compared to the reduced model. This means the stepwise model achieves a better balance between fit and simplicity by removing unnecessary variables.

All remaining variables in the stepwise model are statistically significant at conventional levels (most with p-values < 0.001). This indicates strong evidence that these variables are related to life expectancy.

- Education variables (eys, mys) are significant, showing schooling is an important factor.
- Log GDP and GDP growth are significant, indicating economic conditions matter.
- Health spending is significant, suggesting better healthcare investment improves life expectancy.
- Mortality variables are highly significant and have strong effects, which is expected since they directly relate to health outcomes.
- Population growth is also significant, though its effect is smaller.

Overall, the low p-values suggest that the relationships observed in the model are unlikely to be due to random chance.

In [491…

```
# Check multicollinearity after selection
vif_df_step <- data.frame(Variable=names(vif(step_model)), VIF=vif(step_model)) %>%
dplyr::arrange(desc(VIF))
vif_df_step
```

A data.frame: 8 × 2

	Variable <chr>	VIF <dbl>
mort_inf_f	mort_inf_f	5.780910
log_gdp	log_gdp	4.663775
eys	eys	4.203394
mys	mys	3.793024
mort_adult_f	mort_adult_f	3.457735
health_spending	health_spending	2.362541
population_growth	population_growth	1.437572
GDP.per.capita.growth..annual...	GDP.per.capita.growth..annual...	1.033073

The VIF values are all below approximately 6, indicating that multicollinearity is no longer a major issue. Compared to earlier models, this is a significant improvement, and the predictors now provide more independent information.

In [492...

```
train_control <- trainControl(method="cv", number=5)
cv_model <- train(formula(step_model), data=model_data_clean, method="lm", trContro
cv_model
```

```
Linear Regression

4656 samples
   8 predictor

No pre-processing
Resampling: Cross-Validated (5 fold)
Summary of sample sizes: 3725, 3725, 3725, 3724, 3725
Resampling results:

  RMSE      Rsquared   MAE
  1.587108  0.9673762  1.217216

Tuning parameter 'intercept' was held constant at a value of TRUE
```

Cross-validation results show an RMSE of approximately 1.6 and an R-squared of about 0.967, indicating strong predictive performance. The consistency between cross-validated and training results suggests that the model generalizes well and is not overfitting.

Human Capital Model

In [493...

```
human_model <- lm(life_exp ~ eys + mys + school_enrollment + health_spending +
                  mort_inf_f + mort_adult_f, data=model_data_clean)
summary(human_model)
```

```
Call:
lm(formula = life_exp ~ eys + mys + school_enrollment + health_spending + 
    mort_inf_f + mort_adult_f, data = model_data_clean)

Residuals:
    Min      1Q  Median      3Q     Max 
-7.5415 -1.0767 -0.0438  0.9526  7.9663 

Coefficients:
                    Estimate Std. Error  t value Pr(>|t|)    
(Intercept)       75.7501795  0.2172251  348.717  < 2e-16 ***
eys                0.2392061  0.0175997   13.592  < 2e-16 ***
mys               -0.0426906  0.0152800   -2.794  0.00523 ** 
school_enrollment  0.0052284  0.0018606    2.810  0.00497 ** 
health_spending    0.0036276  0.0001153   31.461  < 2e-16 ***
mort_inf_f        -0.1179330  0.0027119  -43.487  < 2e-16 ***
mort_adult_f      -0.0469820  0.0004614 -101.815  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.763 on 4649 degrees of freedom
Multiple R-squared:  0.9598,	Adjusted R-squared:  0.9597 
F-statistic: 1.848e+04 on 6 and 4649 DF,  p-value: < 2.2e-16
```

In [494…

```
vif_df_human <- data.frame(Variable = names(vif(human_model)), VIF = vif(human_mode
dplyr::arrange(desc(VIF))
vif_df_human
```

A data.frame: 6 × 2

	Variable	VIF
	<chr>	<dbl>
mort_inf_f	mort_inf_f	5.620495
eys	eys	4.891445
school_enrollment	school_enrollment	4.740825
mys	mys	3.731039
mort_adult_f	mort_adult_f	3.496970
health_spending	health_spending	1.898704

The human capital model focuses on education and health variables. The results show that schooling and health spending are positively associated with life expectancy, while mortality variables are negatively associated. The model has strong explanatory power, though slightly lower than the full model.

Economic Model

In [495…

```
economic_model <- lm(life_exp ~ log_gdp + GDP.per.capita.growth..annual... +
                          population_growth, data = model_data_clean)
```

```
summary(economic_model)
```

```
Call:
lm(formula = life_exp ~ log_gdp + GDP.per.capita.growth..annual... +
    population_growth, data = model_data_clean)

Residuals:
    Min      1Q  Median      3Q     Max
-33.426  -2.294   0.698   3.245  13.483

Coefficients:
                                 Estimate Std. Error t value Pr(>|t|)
(Intercept)                      32.68172    0.44500  73.443   <2e-16 ***
log_gdp                           4.53306    0.04882  92.845   <2e-16 ***
GDP.per.capita.growth..annual...  0.01522    0.01351   1.126     0.26
population_growth                -0.60984    0.04762 -12.807   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.901 on 4652 degrees of freedom
Multiple R-squared:  0.6888,	Adjusted R-squared:  0.6886
F-statistic:  3432 on 3 and 4652 DF,  p-value: < 2.2e-16
```

In [496…

```
vif_df_econ <- data.frame(Variable = names(vif(economic_model)), VIF = vif(economic
dplyr::arrange(desc(VIF))
vif_df_econ
```

A data.frame: 3 × 2

	Variable	VIF
	<chr>	<dbl>
population_growth	population_growth	1.108922
log_gdp	log_gdp	1.090549
GDP.per.capita.growth..annual...	GDP.per.capita.growth..annual...	1.027681

The economic model examines income and macroeconomic factors. Log GDP is strongly positively associated with life expectancy, while growth and population effects are smaller. The model explains less variation than the full model, showing that economic factors alone are not sufficient.

Final Model

In [497…

```
final_model <- step_model
summary(final_model)
```

```
Call:
lm(formula = life_exp ~ eys + mys + log_gdp + GDP.per.capita.growth..annual... +
    health_spending + population_growth + mort_adult_f + mort_inf_f,
    data = model_data_clean)

Residuals:
    Min      1Q  Median      3Q     Max
-5.2081 -0.9599 -0.0259  0.9621  6.7442

Coefficients:
                                   Estimate Std. Error  t value Pr(>|t|)
(Intercept)                      69.3382934  0.2778005  249.597  < 2e-16 ***
eys                               0.1490162  0.0146788   10.152  < 2e-16 ***
mys                              -0.1022684  0.0138614   -7.378 1.89e-13 ***
log_gdp                           1.0028108  0.0326794   30.686  < 2e-16 ***
GDP.per.capita.growth..annual... -0.0101887  0.0043854   -2.323   0.0202 *
health_spending                   0.0019264  0.0001157   16.646  < 2e-16 ***
population_growth                 0.0736941  0.0175479    4.200 2.72e-05 ***
mort_adult_f                     -0.0461589  0.0004128 -111.810  < 2e-16 ***
mort_inf_f                       -0.1062953  0.0024745  -42.956  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.586 on 4647 degrees of freedom
Multiple R-squared:  0.9674,	Adjusted R-squared:  0.9674
F-statistic: 1.726e+04 on 8 and 4647 DF,  p-value: < 2.2e-16
```

In [498…

```
vif_df_final <- data.frame(Variable=names(vif(final_model)), VIF=vif(final_model))
dplyr::arrange(desc(VIF))
vif_df_final
```

A data.frame: 8 × 2

	Variable	VIF
	<chr>	<dbl>
mort_inf_f	mort_inf_f	5.780910
log_gdp	log_gdp	4.663775
eys	eys	4.203394
mys	mys	3.793024
mort_adult_f	mort_adult_f	3.457735
health_spending	health_spending	2.362541
population_growth	population_growth	1.437572
GDP.per.capita.growth..annual...	GDP.per.capita.growth..annual...	1.033073

The final model is based on the stepwise selection results. It includes variables related to education (eys, mys), economic conditions (log_gdp, GDP growth), health spending, population growth, and mortality.

The model achieves an adjusted R-squared of approximately 0.967, indicating strong explanatory power. All variables are statistically significant, suggesting meaningful relationships with life expectancy.

Multicollinearity has been reduced, with all VIF values below acceptable thresholds, making the model more stable and interpretable.

Cross-validation was performed using 5-fold validation. The model achieved an RMSE of approximately 1.6 and an R-squared of about 0.967.

These results indicate strong predictive performance. The similarity between training and validation results suggests that the model generalizes well and is not overfitting.

Predicted vs Actual Plot - Final Model

In [499…

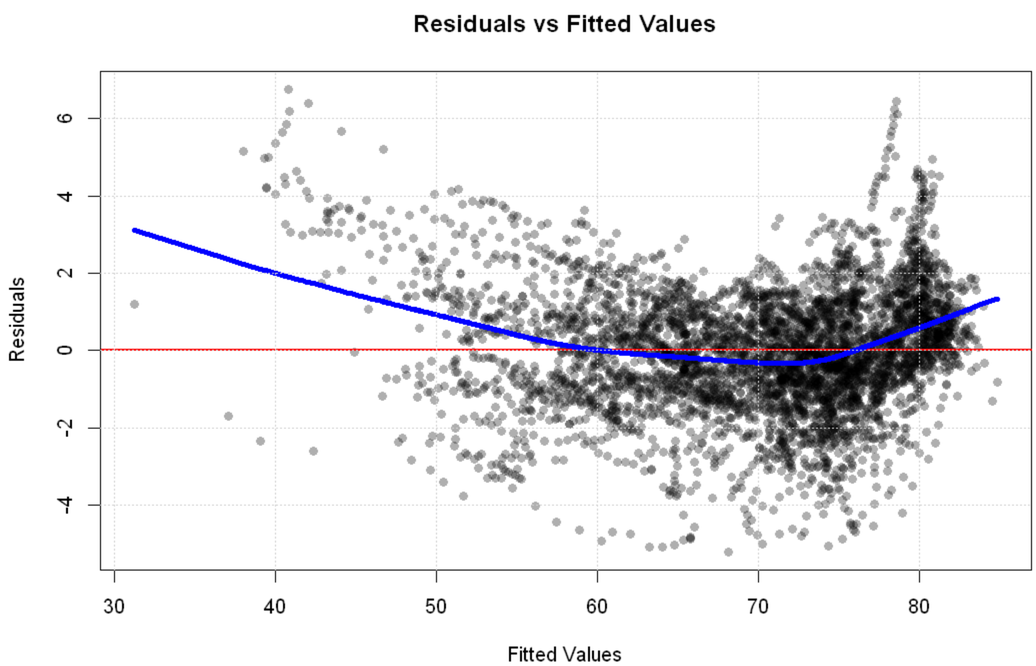
```
plot(model_data_clean$life_exp, pred,
     pch=16, col=rgb(0,0,0,0.3),
     xla="Actual Life Expectancy",
     ylab="Predicted Life Expectancy",
     main="Predicted vs Actual Life Expectancy")

abline(0, 1, col="red", lwd = 2)
grid()
```

The predicted vs actual plot shows how closely the model's predictions match the observed life expectancy values. Most points lie very close to the 45-degree line, indicating that the model produces accurate predictions across countries. There is slightly more spread at lower life expectancy levels, suggesting the model is less precise for lower-income or higher-mortality countries. Overall, the tight clustering around the line confirms strong model fit and high predictive accuracy

Residuals vs Fitted Plot - Final Model

In [500…

```
plot(final_model$fitted.values, resid(final_model),
     pch=16, col=rgb(0,0,0,0.3),
     xlab="Fitted Values",
     ylab="Residuals",
     main="Residuals vs Fitted Values")

abline(h=0, col="red", lwd = 2)
lines(lowess(final_model$fitted.values, resid(final_model)), col="blue", lwd = 4)

grid()
```



The residuals vs fitted plot is used to assess model assumptions. The residuals are generally centered around zero, which supports the linearity assumption. However, there is some visible curvature and increasing spread at higher fitted values, indicating mild heteroskedasticity, which could be addressed in further analyses

Key Findings

- Higher education levels are associated with higher life expectancy
- Higher GDP is associated with improved health outcomes
- Health spending has a positive effect
- Mortality rates have strong negative effects
- Population growth has a smaller but significant impact

Summary of Analysis

This analysis investigates the main factors influencing life expectancy using a panel dataset constructed from WHO, World Bank, and UNDP data. The initial regression model showed very strong explanatory power but also revealed high multicollinearity, particularly among mortality, survival, and HDI-related variables, which made interpretation less reliable.

To address this, influential observations were identified using Cook's Distance and a small number of extreme outliers were removed. The model was then simplified by eliminating redundant variables, which reduced multicollinearity to acceptable levels. Stepwise AIC was applied to further refine the model, resulting in a more parsimonious specification without a significant loss in fit.

The final model explains a large proportion of the variation in life expectancy (adjusted $R^2 \approx 0.967$), and cross-validation results ($RMSE \approx 1.6$) indicate that the model generalizes well to new data.

Overall, the results show that higher levels of education and income are associated with increased life expectancy, while higher mortality rates are strongly associated with lower life expectancy. Health spending also has a positive and statistically significant effect, and population growth shows a smaller but meaningful relationship.

In conclusion, life expectancy appears to be driven by a combination of economic development, education, and health outcomes, with mortality variables having the strongest influence in the model.